

MOHAMED AZHARUDEEN K

Senior Software Engineer

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SUMMARY

Senior Automotive Embedded Software Engineer with 7+ years of experience in AUTOSAR Architecture and Embedded Development, specializing in BSW, RTOS, CAN/CAN-FD communication. Proficient in Embedded C, DaVinci tools, CANoe, and TKWinx with strong expertise in requirements analysis, Software debugging and ASPICE-compliant development aligned with ISO 26262 standards. Proven track record of delivering reliable automotive solutions for OEM programs and improving software quality through effective defect reduction and testing practices.

SKILLS

Languages:	C Programming, Embedded C, C++, Python
Development Tools:	SharCC, AEEE_Pro, DaVinci Configurator, STM32 Cube IDE, MPLAB
Protocols:	CAN, UDS, UART, I2C, SPI
Testing Tools:	Vector CANoe/CANalyzer, TKWinx, DET Tool, Cantata, MAVS
Debug Tools:	JTAG, Oscilloscope, Lauterbach, Vehicle Network Tester (IVNT-03)
Microcontroller:	Renesas RH850, STM32 Series, ARM7-LPC2148, PIC16F8XX
Automotive Standards:	ASPICE, AUTOSAR, ISO 26262, ISO 11898
Soft Skills:	Design Understanding, Team Collaboration, Quality Focus, Adaptability

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Bosch Global Software Technologies | Coimbatore | 2nd Aug 2021 – 18th Mar 2026

1. Project – Ford Commercial Vehicle – V710ESP, V710IPB & V363ICA3

- Configured and integrated COM stack module using Davinci configurator , AEE Pro and SharCC IDE
- Implemented various Tx/Rx messages and signals based on the customer requirements
- Performed DBC file analysis and updates by comparing new customer-provided DBCs with existing versions, identifying changes in CAN messages and signals
- Ensured reliable CAN communication by implementing and validating checksum and rolling counter mechanisms for error detection and data integrity
- Developed software changes and validated them through Hardware-in-the-Loop (HIL) testing to ensure functional correctness and requirement compliance.
- Delivered project tasks with an average of 98% on-time completion rate, demonstrating strong planning, ownership, and adherence to timelines
- Hands-on experience in setting up and configuring test benches for component testing
- Preparing comprehensive technical documentation, including design specifications, user manuals, and test plans, to facilitate knowledge transfer and support future development efforts
- Strong experience in automotive SDLC following the V-Model, including requirements analysis, software design, implementation, Unit testing, and verification/validation

2. Project – Ford Hybrid Passenger Vehicle – B2E IPB

- Configured communication parameters for CAN, CAN FD, and XCP on Can protocols to enable reliable data exchange across ECUs.
- Implemented and validated XCP protocol for ECU calibration and measurement activities
- Developed a diagnostic-based XCP activation counter (0–255), integrated with Diagnostic Tools (DET) via DCOM DID allowing for controlled, temporary access to ECU calibration and data acquisition
- Led the Defect Reduction Process (DRP) for network components, performed root cause analysis (RCA), and achieved a 37% reduction in defects within a calendar year

- Implemented message timeout handling with DEM fault reporting for missing CAN messages
- Experienced in AUTOSAR CAN Device Driver configuration (MCAL), including controller setup, baud rate configuration to enable reliable communication between ECU hardware and software layers
- Utilizing advanced debugging tools and techniques to troubleshoot and resolve complex software issues, enhancing system reliability and performance
- Proficiency in quality assurance and handling testing tools like CANoe, TKWinx, DET Tool, and MAVS, Micro Focus ALM with hands-on experience in HIL and SIL Testing
- Execution of unit testing of software changes using Cantata to achieve the traceability, compliance, and quality

3. Project – Ford EU7 Hybrid Passenger Vehicle – BX726 & Ford - F150

- Designed and implemented Network Management (NM) features ensuring accurate ECU state transitions and optimized communication behavior
- Mapping of NM state information to DIDs, enabling diagnostic tools to read ECU network status
- Configured and implemented Standard CAN (11-bit ID) and Extended CAN (29-bit ID) messages
- Experience in regression testing to ensure stability of existing functionalities after software changes
- Ability to create and maintain functional design documents for software requirements
- Configured NVM blocks for data storage and retrieval across ignition cycles
- Good experience in applying ASPICE compliance to ensure high-quality software development for automotive systems aligned with ISO 26262 functional safety guidelines
- Analyzed static reports to address MISRA violations, errata, TPA, and compiler warnings
- Experienced in using GitHub Copilot for AI-assisted software development and improved development efficiency by utilizing intelligent code suggestions and automation support
- Prepared software design using Enterprise Architect (EA) based on functional requirements
- Analyzed and understood customer requirements and documented those using IBM Rational DOORS, ensuring end-to-end traceability and alignment with development

Embedded Developer | Arris Automation & Solutions | Coimbatore | 14th May 2018 – 23rd July 2021

- Designed and developed embedded applications using development tools like STM32 CubeIDE and MPLAB
- Analyzed microcontroller pin configurations, schematics, and datasheets to support seamless hardware-software integration
- Proficient in C and Embedded C programming, with a rich understanding of microcontrollers
- Implemented ADC, PWM, and GPIO functionalities to enable accurate sensor readings, actuator control, and digital interfacing in embedded systems
- Developed IoT projects with sensor integration for real-time monitoring and automation
- Designed and implemented I2C, SPI, and UART interfaces for microcontroller-based systems, ensuring efficient and reliable communication
- Hands-on experience with wireless and IoT networks like Bluetooth, ESP8266, Raspberry Pi

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering | Aug 2014 – April 2018
 SNS College of Technology, Coimbatore | CGPA: 8.64 | First Class with Distinction

CERTIFICATIONS

- Failure Mode Effect Analysis (FMEA) – Bosch Global Software Technologies
 - Master CAN protocol completely from Scratch (CAN + CAN-FD) – Udemy
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AWARDS

Received the **“Bravo Award”** from Group Manager for contribution to development, efficiency, and team collaboration at Bosch Global Software Technologies